

IMO (56th) – 2015

First Day

1. We say that a finite set S of points in the plane is balanced if, for any two different points A and B in S , there is a point C in S such that $AC = BC$. We say that S is centre-free if for any three different points A , B and C in S , there is no point P in S such that $PA = PB = PC$.
 - (a) Show that for all integers $n \geq 3$, there exists a balanced set consisting of n point.
 - (b) Determine all integers $n \geq 3$ for which there exists a balanced centre-free set consisting of n points.
2. Determine all triples (a, b, c) of positive integers such that each of the numbers $ab - c, bc - a, ca - b$ is a power of 2. (A power of 2 is an integer of the form 2^n , where n is a non-negative integer.)
3. Let ABC be an acute triangle with $AB > AC$. Let Γ be its circum circle, H its orthocenter, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$, and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A , B , C , K and Q are all different, and lie on Γ in this order. Prove that the circumcircles of triangles KQH and FKM are tangent to each other.

Second Day

4. Triangle ABC has circumcircle Ω and circumcentre O . A circle Γ with centre A intersects the segment BC at points D and E , such that B, D, E and C are all different and lie on line BC in this order. Let F and G be the points of intersection of Γ and Ω , such that A, F, B, C and G lie on Ω in this order. Let K be the second point of intersection of the circumcircle of triangle BDF and the segment AB . Let L be the second point of the intersection of the circumcircle of triangle CGE and the segment CA . Suppose that the lines FK and GL are different and intersect at the point X . Prove that X lies on the line AO .
5. Let $\mathfrak{R}$ be the set of real numbers. Determine all functions $f : \mathfrak{R} \rightarrow \mathfrak{R}$ satisfying the equation $f(x + f(x + y)) + f(xy) = x + f(x + y) + yf(x)$ for all real numbers x and y .
6. The sequence $a_1, a_2, \dots$ of integers satisfies the following conditions:
 - (a) $1 \leq a_j \leq 2015$ For all $j \geq 1$.
 - (b) $k + a_k \neq l + a_l$ For all $1 \leq k \leq l$.

Prove that there exist two positive integers b and N such that $\left| \sum_{j=m+1}^n (a_j - b) \right| \leq 1007^2$ for

all integers m and n satisfying $n > m \geq N$.
